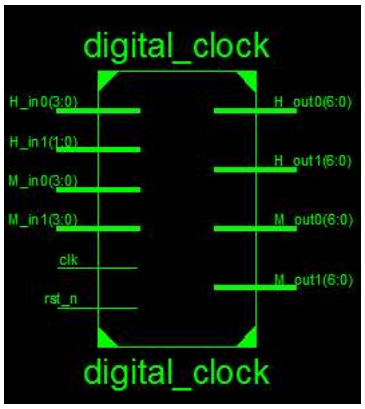
VHDL code for digital clock on FPGA

This VHDL project is the VHDL version code of the digital clock in Verilog I posted before(link). The VHDL code for the digital clock is synthesizable for FPGA implementation and full VHDL code is provided.

This digital clock is a reconfigurable 24-hour clock displaying hours, minutes, and seconds on seven-segment LEDs (Tutorials on 7-segment LEDs: here). Besides, users can manually set the time of the digital clock including hours and minutes through switches.



VHDL code for digital clock:

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
use IEEE.numeric_std.all;
-- fpga4student.com FPGA projects, VHDL projects, Verilog projects
-- VHDL project: VHDL code for digital clock
entity digital_clock is
port (
 clk: in std_logic;
 -- clock 50 MHz
 rst_n: in std_logic;
 -- Active low reset pulse, to set the time to the input hour and minute (as
 -- defined by the H_in1, H_in0, M_in1, and M_in0 inputs) and the second to 00.
 -- For normal operation, this input pin should be 1.
 H_in1: in std_logic_vector(1 downto 0);
 -- 2-bit input used to set the most significant hour digit of the clock
 -- Valid values are 0 to 2.
 H_in0: in std_logic_vector(3 downto 0);
 -- 4-bit input used to set the least significant hour digit of the clock
 -- Valid values are 0 to 9.
 M_in1: in std_logic_vector(3 downto 0);
 -- 4-bit input used to set the most significant minute digit of the clock
 -- Valid values are 0 to 9.
 M_in0: in std_logic_vector(3 downto 0);
 -- 4-bit input used to set the least significant minute digit of the clock
 -- Valid values are 0 to 9.
 H_out1: out std_logic_vector(6 downto 0);
 -- The most significant digit of the hour. Valid values are 0 to 2 ( Hexadecimal
 H_out0: out std_logic_vector(6 downto 0);
 -- The most significant digit of the hour. Valid values are 0 to 9 ( Hexadecimal
 M_out1: out std_logic_vector(6 downto 0);
 -- The most significant digit of the minute. Valid values are 0 to 9 ( Hexadecima
 M_out0: out std_logic_vector(6 downto 0)
 -- The most significant digit of the minute. Valid values are 0 to 9 ( Hexadecima
 );
end digital_clock;
architecture Behavioral of digital_clock is
component bin2hex
```

```
port (
 Bin: in std_logic_vector(3 downto 0);
 Hout: out std_logic_vector(6 downto 0)
 );
end component;
component clk_div
port (
 clk_50: in std_logic;
 clk_1s: out std_logic
     );
end component;
-- fpga4student.com FPGA projects, VHDL projects, Verilog projects
signal clk_1s: std_logic; -- 1-s clock
signal counter_hour, counter_minute, counter_second: integer;
-- counter using for create time
signal H_out1_bin: std_logic_vector(3 downto 0); --The most significant digit of t
signal H_out0_bin: std_logic_vector(3 downto 0);--The least significant digit of t
signal M_out1_bin: std_logic_vector(3 downto 0);--The most significant digit of th
signal M_out0_bin: std_logic_vector(3 downto 0);--The least significant digit of t
begin
-- create 1-s clock --|
create_1s_clock: clk_div port map (clk_50 => clk, clk_1s => clk_1s);
-- clock operation ---|
process(clk_1s,rst_n) begin
-- fpga4student.com FPGA projects, VHDL projects, Verilog projects
  if(rst_n = '0') then
 counter_hour <= to_integer(unsigned(H_in1))*10 + to_integer(unsigned(H_in0));
 counter_minute <= to_integer(unsigned(M_in1))*10 + to_integer(unsigned(M_in0));
 counter_second <= 0;
 elsif(rising_edge(clk_1s)) then
 counter_second <= counter_second + 1;
 if(counter_second >=59) then -- second > 59 then minute increases
 counter_minute <= counter_minute + 1;
 counter_second <= 0;
 if(counter_minute >=59) then -- minute > 59 then hour increases
 counter_minute <= 0;
 counter_hour <= counter_hour + 1;
 if(counter_hour >= 24) then -- hour > 24 then set hour to 0
 counter_hour <= 0;
 end if;
 end if;
 end if;
 end if;
end process;
----------------------|
-- Conversion time ---|
----------------------|
-- H_out1 binary value
 H_out1_bin <= x"2" when counter_hour >=20 else
 x"1" when counter_hour >=10 else
 x"0";
-- 7-Segment LED display of H_out1
convert_hex_H_out1: bin2hex port map (Bin => H_out1_bin, Hout => H_out1);
-- H_out0 binary value
 H_out0_bin <= std_logic_vector(to_unsigned((counter_hour - to_integer(unsigned(H_
-- 7-Segment LED display of H_out0
convert_hex_H_out0: bin2hex port map (Bin => H_out0_bin, Hout => H_out0);
-- M_out1 binary value
 M_out1_bin <= x"5" when counter_minute >=50 else
 x"4" when counter_minute >=40 else
 x"3" when counter_minute >=30 else
 x"2" when counter_minute >=20 else
 x"1" when counter_minute >=10 else
```

```
 x"0";
-- 7-Segment LED display of M_out1
convert_hex_M_out1: bin2hex port map (Bin => M_out1_bin, Hout => M_out1);
-- M_out0 binary value
 M_out0_bin <= std_logic_vector(to_unsigned((counter_minute - to_integer(unsigned(
-- 7-Segment LED display of M_out0
convert_hex_M_out0: bin2hex port map (Bin => M_out0_bin, Hout => M_out0);
end Behavioral;
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-- VHDL project: VHDL code for digital clock
-- BCD to HEX For 7-segment LEDs display
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
entity bin2hex is
port (
 Bin: in std_logic_vector(3 downto 0);
 Hout: out std_logic_vector(6 downto 0)
);
end bin2hex;
architecture Behavioral of bin2hex is
begin
 process(Bin)
 begin
  case(Bin) is
   when "0000" =>  Hout <= "1000000"; --0--
   when "0001" =>  Hout <= "1111001"; --1--
   when "0010" =>  Hout <= "0100100"; --2--
   when "0011" =>  Hout <= "0110000"; --3--
   when "0100" =>  Hout <= "0011001"; --4--
   when "0101" =>  Hout <= "0010010"; --5--
   when "0110" =>  Hout <= "0000010"; --6--
   when "0111" =>  Hout <= "1111000"; --7--
   when "1000" =>  Hout <= "0000000"; --8--
   when "1001" =>  Hout <= "0010000"; --9--
   when "1010" =>  Hout <= "0001000"; --a--
   when "1011" =>  Hout <= "0000011"; --b--
   when "1100" =>  Hout <= "1000110"; --c--
   when "1101" =>  Hout <= "0100001"; --d--
   when "1110" =>  Hout <= "0000110"; --e--
   when others =>  Hout <= "0001110";
   end case;
 end process;
end Behavioral;
-- fpga4student.com FPGA projects, VHDL projects, Verilog projects
-- VHDL project: VHDL code for digital clock
-- Clock divider module to get 1 second clock
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
USE IEEE.STD_LOGIC_UNSIGNED.ALL;
entity clk_div is
port (
   clk_50: in std_logic;
   clk_1s: out std_logic
  );
end clk_div;
architecture Behavioral of clk_div is
signal counter: std_logic_vector(27 downto 0):=(others =>'0');
begin
 process(clk_50)
 begin
  if(rising_edge(clk_50)) then
   counter <= counter + x"0000001";
   if(counter>=x"2FAF080") then -- for running on FPGA -- comment when running sim
```

```
   --if(counter_slow>=x"0000001") then -- for running simulation -- comment when r
    counter <= x"0000000";
   end if;
  end if;
 end process;
 clk_1s <= '0' when counter < x"17D7840" else '1';
end Behavioral;
```

Testbench VHDL code for digital clock:

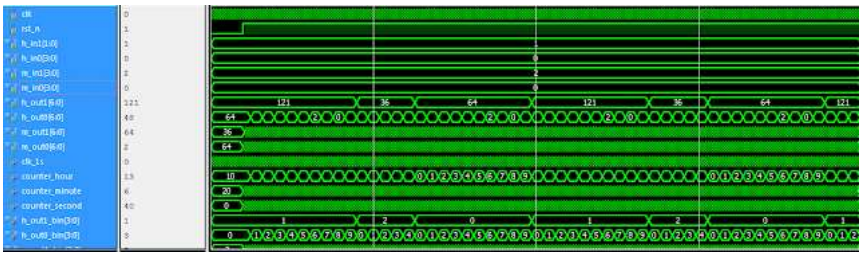
```
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;
-- fpga4student.com FPGA projects, VHDL projects, Verilog projects
-- VHDL project: VHDL code for digital clock
ENTITY tb_digital_clock IS
END tb_digital_clock;
ARCHITECTURE behavior OF tb_digital_clock IS
  -- Component Declaration for the Unit Under Test (UUT)
-- fpga4student.com FPGA projects, VHDL projects, Verilog projects
    COMPONENT digital_clock
    PORT(
         clk : IN  std_logic;
         rst_n : IN  std_logic;
         H_in1 : IN  std_logic_vector(1 downto 0);
         H_in0 : IN  std_logic_vector(3 downto 0);
         M_in1 : IN  std_logic_vector(3 downto 0);
         M_in0 : IN  std_logic_vector(3 downto 0);
         H_out1 : OUT  std_logic_vector(6 downto 0);
         H_out0 : OUT  std_logic_vector(6 downto 0);
         M_out1 : OUT  std_logic_vector(6 downto 0);
         M_out0 : OUT  std_logic_vector(6 downto 0)
        );
    END COMPONENT;
   --Inputs
   signal clk : std_logic := '0';
   signal rst_n : std_logic := '0';
   signal H_in1 : std_logic_vector(1 downto 0) := (others => '0');
   signal H_in0 : std_logic_vector(3 downto 0) := (others => '0');
   signal M_in1 : std_logic_vector(3 downto 0) := (others => '0');
   signal M_in0 : std_logic_vector(3 downto 0) := (others => '0');

  --Outputs
   signal H_out1 : std_logic_vector(6 downto 0);
   signal H_out0 : std_logic_vector(6 downto 0);
   signal M_out1 : std_logic_vector(6 downto 0);
   signal M_out0 : std_logic_vector(6 downto 0);
   -- Clock period definitions
   constant clk_period : time := 10 ps;
BEGIN
-- fpga4student.com FPGA projects, VHDL projects, Verilog projects
 -- Instantiate the Unit Under Test (UUT)
   uut: digital_clock PORT MAP (
          clk => clk,
          rst_n => rst_n,
          H_in1 => H_in1,
          H_in0 => H_in0,
          M_in1 => M_in1,
          M_in0 => M_in0,
          H_out1 => H_out1,
          H_out0 => H_out0,
          M_out1 => M_out1,
          M_out0 => M_out0
        );
   -- Clock process definitions
```

```
   clk_process :process
   begin
 clk <= '0';
 wait for clk_period/2;
 clk <= '1';
 wait for clk_period/2;
   end process;
   -- Stimulus process
   stim_proc: process
   begin
      -- hold reset state for 100 ns.
 rst_n <= '0';
 H_in1 <= "01";
 H_in0 <= x"0";
 M_in1 <= x"2";
 M_in0 <= x"0";
      wait for 100 ns;
 rst_n <= '1';
      wait for clk_period*10;
-- insert stimulus here
-- fpga4student.com FPGA projects, VHDL projects, Verilog projects
      wait;
   end process;

END;
```

Simulation results for the digital clock in VHDL:



It is noted that when you run the simulation, you should use direct clock from input to simulate instead of the 1-second clock or reduce the value of the divisor in the clk_div entity because it would take a long time to create the 1-second clock.

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